VERSION
14.0



Software Architecture

AFLCMC/WA

April 24, 2026

AMS GRA ARCHITECTURE VOLUMES

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1.0 Overview

The Software Architecture defines the framework for the development and deployment of software onto an Agile Mission Suite (AMS) Government Reference Architecture (GRA)-compliant system.

1.1 Purpose

The Software Architecture Volume defines and describes the software architecture elements used in AMS GRA as well as the relationships between them. This Software Architecture Volume also defines Development Kits (DKs) required for some Minimum Procurable Units (MPUs) to allow for the development of software.

1.2 Scope

The Software Architecture Volume aids Level 1 MPU developers in understanding how Level 1 MPUs interact with other elements in the AMS GRA. This volume is also applicable to entities required to produce a DK while performing an AMS GRA compliance assessment. This volume is not prescriptive in that it enumerates no AMS GRA compliance requirements. The delineation between MPU levels and the identification of which AMS GRA Elements are Level 1 MPUs can be found in the AMS GRA Mission System Volume. Additionally, each Level 1 MPU has an associated architecture volume with specific compliance guidance.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, Mission System Integrator (MSI), or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. See **Table 1.3.1** for a list of authoritative references for the Software Architecture Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	PCaB Compliance Documentation Template	14.0
AMS GRA Models	AMS GRA Architecture Model	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of the Software Architecture Volume.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the Software Architecture Volume.

Standard	Title	Version/Date
AMS GRA Volumes	AMS GRA Service MPU Volume	14.0
AMS GRA Volumes	Assembly MPU Volume	14.0
AMS GRA Volumes	CDS IDD	14.0
AMS GRA Volumes	Cyber IDD	14.0
AMS GRA Volumes	DPP MPU Volume	14.0
AMS GRA Volumes	IR MEL IDD	14.0
AMS GRA Volumes	Local Function MPU Volume	14.0
AMS GRA Volumes	MASI Volume	14.0
AMS GRA Volumes	MFA MPU Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	OMS Service MPU Volume	14.0
AMS GRA Volumes	RF MEL IDD	14.0
OMS Standard	OMS Description and Documentation	2.3

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the Software Architecture Volume and Related Entities.

2.0 Software Framework

Figure 2.0.1 depicts the Software Framework for the development and deployment of software onto an AMS GRA mission system. The Mission Agnostic Service Infrastructure (MASI) and Skills are software developed using DKs and deployed in the AMS GRA Compute Environment. Development, Security, and Operations (DevSecOps) enables the ability to adapt to changes and rapidly realign based on customer need or new opportunities, all without a decrease in efficiency. The set of Mission Specific and Mission Agnostic Skills depicted in **Figure 2.0.1** is notional.

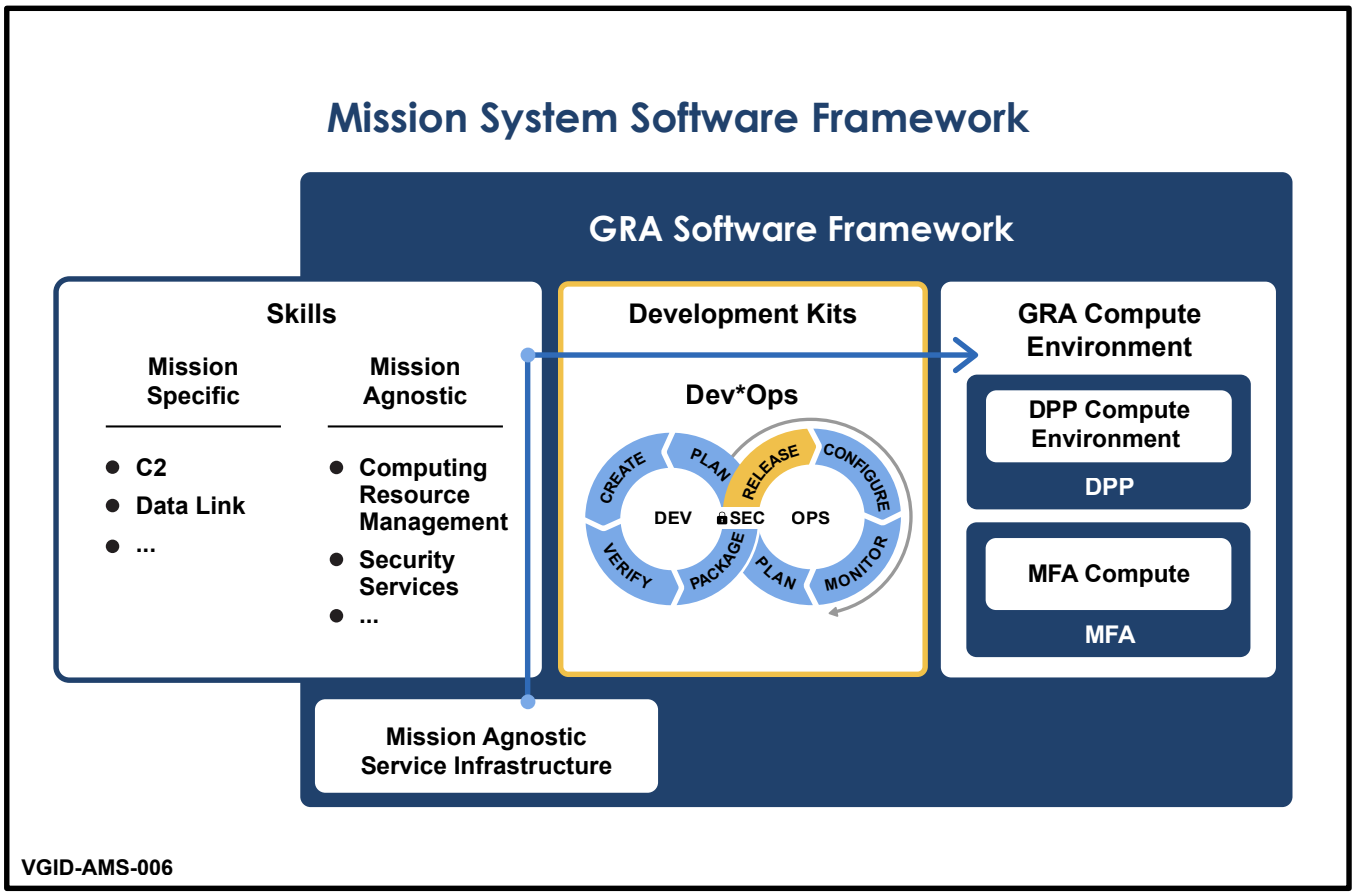


Figure 2.0.1: The AMS GRA Mission System Software Framework Leverages: the MASI, DKs, and Strong DevSecOps Principles to Rapidly Develop and Deploy Software in Alignment with Customer Needs.

Figure 2.0.1 shows the relationship between a Mission System Software Framework and the AMS GRA Software Framework. In particular, a Mission System Software Framework leverages the AMS GRA Software Framework. Whereas a Mission System Software Framework is tailored to a specific set of missions, the AMS GRA Software Framework is not. The AMS GRA Software Framework and its instantiations are mission agnostic for greater reusability.

The AMS GRA Software Framework includes the following elements: Software Architecture, Development Environment, and Compute Environment as shown in **Figure 2.0.2**. The remainder of the Software Architecture Volume addresses these elements and software interfaces.

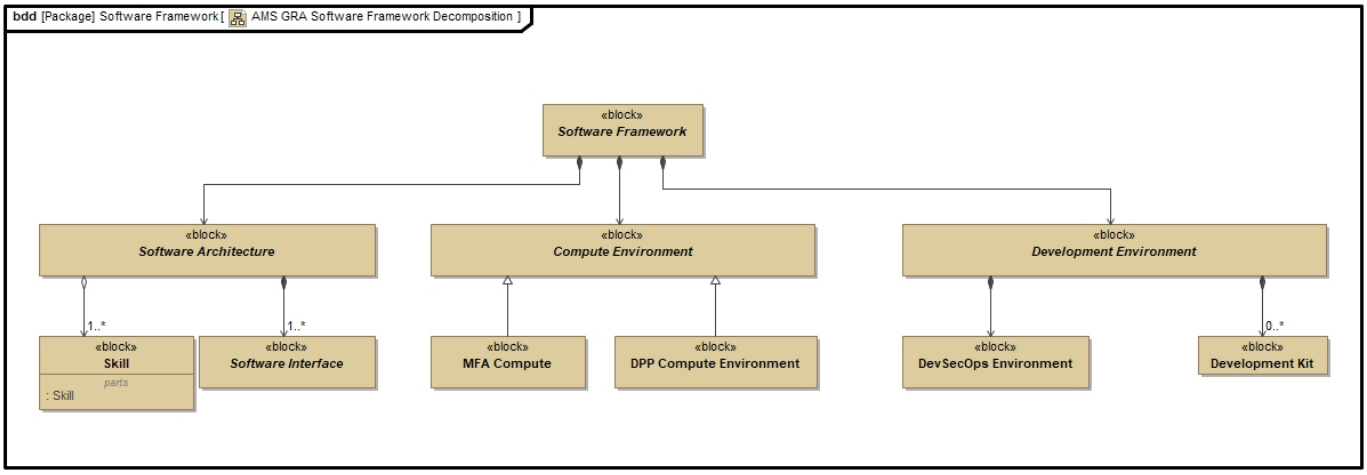


Figure 2.0.2: The AMS GRA Software Framework is Composed of the Software Architecture, Development Environment, and the Compute Environment.

2.1 Software Architecture

2.1.1 Skills

A Skill is a procurable unit that provides a desired effect for the Mission System through software. Skills, while procurable, are not MPUs. Skills are an aggregation of Service and Local Function MPUs (as shown in **Figure 2.1.1**).

As shown in **Figure 2.0.1**, Skills are either Mission Specific or Mission Agnostic. Mission Agnostic Skills may be provided with an AMS GRA Software Framework implementation, whereas Mission Specific Skills are not provided

with the AMS GRA Software Framework implementation. Mission Agnostic Skills are mission-invariant and may support multiple other Skills in a standard way. Mission Specific Skills add mission capability to an AMS GRA Mission System instantiation and likely vary between AMS GRA Mission System instantiations.

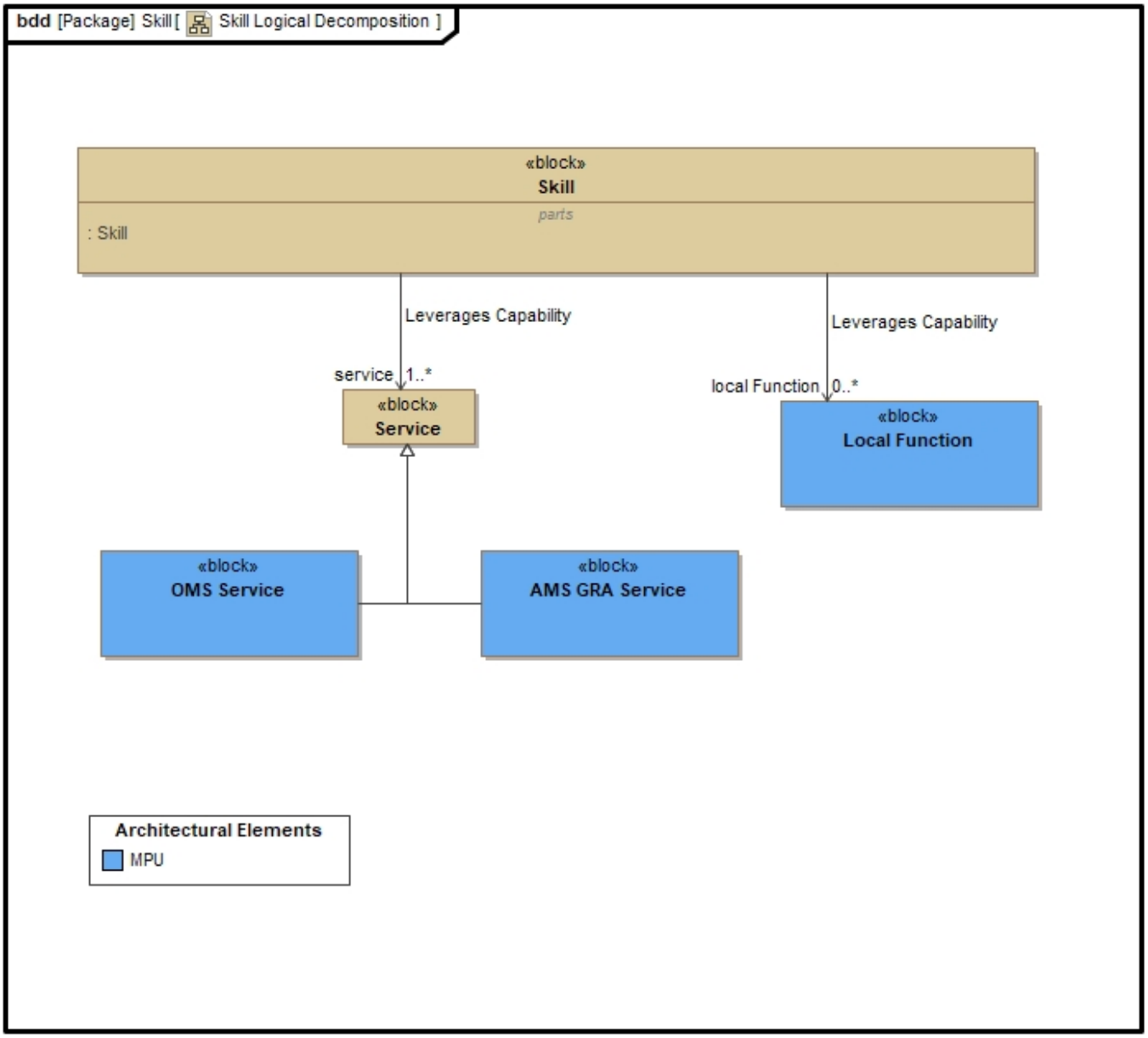


Figure 2.1.1: Each AMS GRA Skill is Made Up of One or More Software MPUs (Local Function, OMS Service, and/or AMS GRA Service). This Modular Approach Allows the AMS GRA to Take Advantage of Reuse and Speed of Development.

Figure 2.1.1 introduces the Digital Payload (DP). A DP is the smallest unit of deployable software on a single Digital Payload Processor (DPP). A Service is composed of one or more DPs, and at least one DP must expose the Service's Open interface. When multiple DPs compose a Service, those DPs may be co-located on the same DPP or distributed across multiple DPPs, and the DPs may communicate to each other using internal, non-exposed interfaces. **Figure 2.1.2** visualizes the allowable composition and deployment of DPs within Services on DPPs.

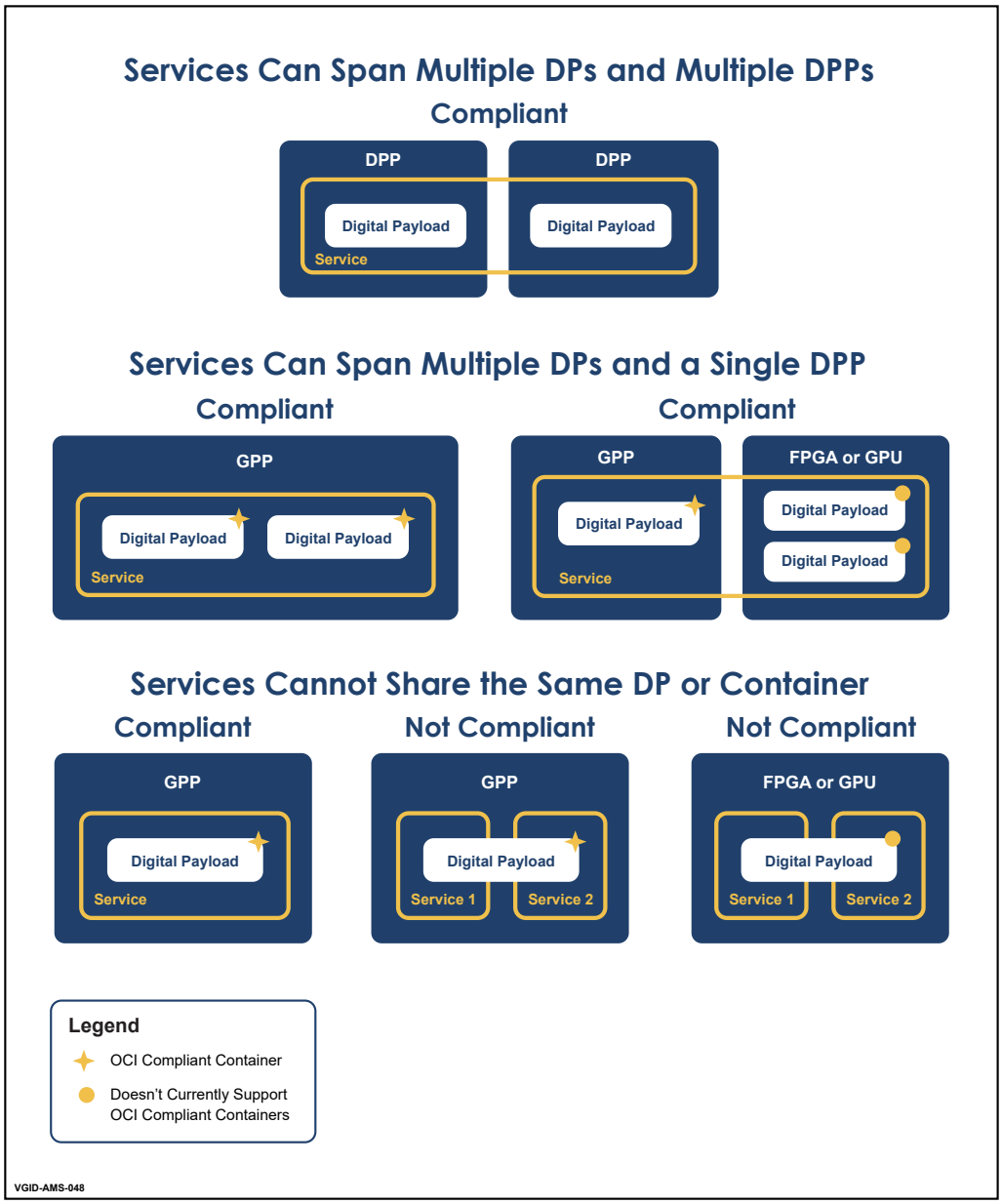


Figure 2.1.2: Software Services May be Deployed in Either Single or Multiple DPPs Configurations, so Long as Each Individual DP Resides Only on a Single DPP and Within a Single Service.

A Service boundary is defined as where the Service communications enter or exit the Service. All exchanges that cross a Service boundary must be fully documented in its non-proprietary Service Contract.

Figure 2.1.3 illustrates a few of the potential aggregations of MPUs in a Skill. Skills may exist on either a single layer or span across Mission Processing Subsystem (MPS), Multi-Function Processor (MFP), and Multi-Function Aperture (MFA) layers.

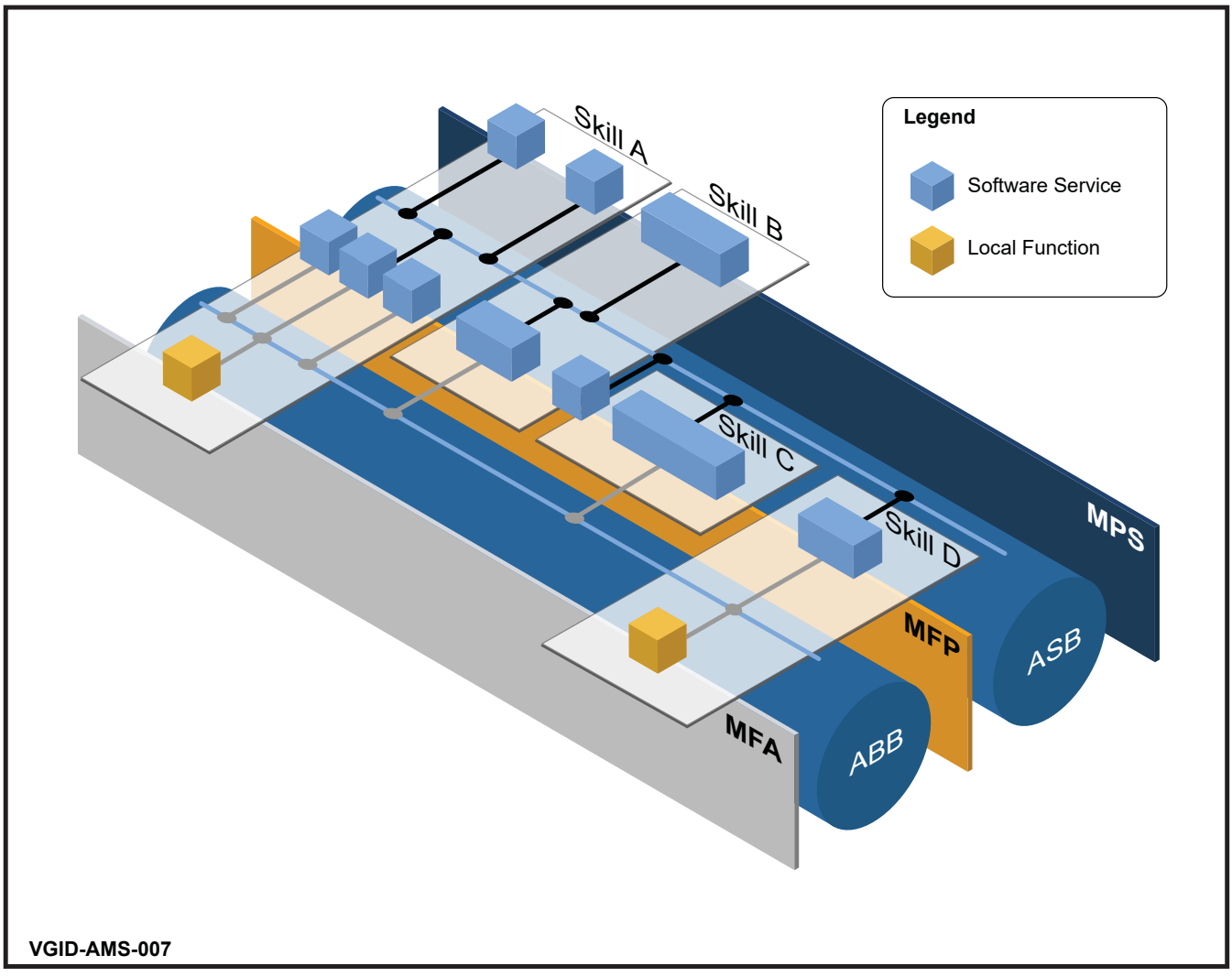


Figure 2.1.3: An AMS GRA Skill May Cross the MPS, MFP, and MFA Layers so Long as Its Component Services and Local Functions Individually Reside on One Single Layer.

In **Figure 2.1.3**, Skill A demonstrates a Skill that crosses the MPS, MFP, and MFA layers. This Skill consists of Services as well as a Local Function in the MFA layer. Skill B represents a Skill that has Services existing at both the MPS and MFP layers. Skill C represents a Skill that exists in a single layer–in this case the MFP layer. Skills B and C demonstrate that a Service (shown as spanning the boundary between the two Skills in the MFP layer) can be shared by multiple Skills. The Skill aggregations, shown in **Figure 2.1.3**, are all valid options.

2.1.2 Software Interfaces

The Software Interfaces defined by the AMS GRA Software Framework are described in the Interfaces Table of the AMS GRA Volume that corresponds to the MPU that provides each interface, and in some cases the accompanying Interface Description Document (IDD). The remainder of **Section 2.1.2** provides Software Interface information not found in other AMS GRA Volumes or IDDs.

Figure 2.1.4 depicts the deployment of three Services (Open Mission Systems (OMS) Service A, OMS Service B, and AMS GRA Service A) to an AMS GRA aligned Compute Cluster Farm (CCF) hardware configuration. The General Purpose Processors (GPPs), AMS GRA Software Framework elements, and Interfaces identified within the figure are only one potential configuration, rather than an exhaustive depiction.

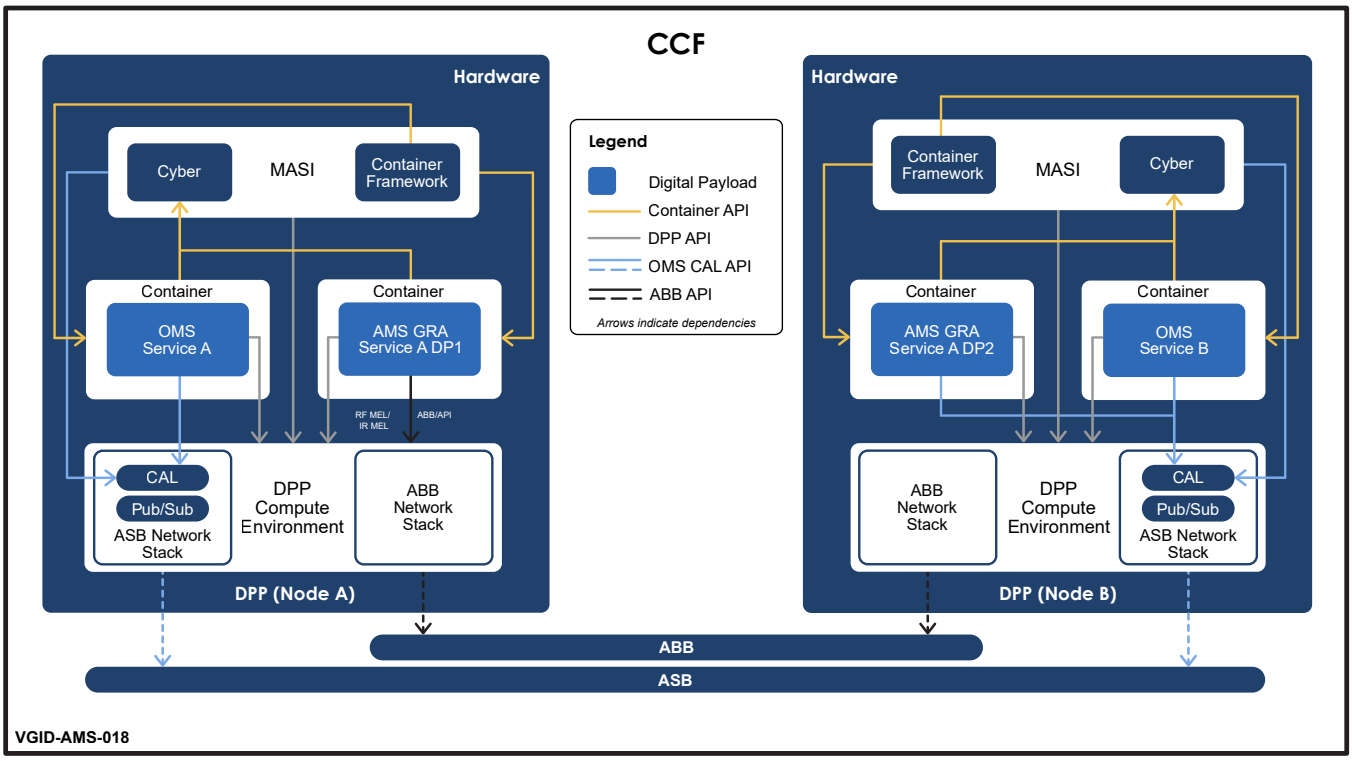


Figure 2.1.4: The AMS GRA Software Architecture Allows for Multiple Possible Deployment Configurations of Framework Elements, Allowing Adopters to Choose the Configuration that Best Aligns with Their Mission or Platform Needs.

2.1.2.1 DPP Application Programming Interface (API)

The DPP API aggregates the interfaces available to AMS GRA and OMS Services on a DPP to interact with GPP, Field-Programmable Gate Array (FPGA), and Graphics Processing Unit (GPU) Processing Components.

the GPP DPP API uses the Portable Operating System Interface (POSIX) for Operating System (OS) system calls.

Of the three DPP types (FPGA, GPU, GPP), only the FPGA API is entirely defined by the DPP DK. The GPU and GPP DPP APIs have some API definitions required by the DPP DK, but may have additional API requirements placed on one or both of them by the MSI. An example of an additional API that an MSI could leverage would be mandating OpenGL support.

2.1.2.2 Avionics Backplane Bus (ABB) Interface

An ABB API is an interface used by software to exchange data over the ABB. ABB data may be user-defined, but must adhere to the openness requirements on the communicating elements. Open Source Software (OSS)/Commercial-Off-The-Shelf (COTS) ABB APIs (and associated implementation libraries) are used to provide the desirable ABB data exchange patterns:

- Send/Receive (Network Packets)
- Publisher/Subscriber
- Point-to-Point
- Streaming
- Client/Server
- Direct Memory Transfer (e.g., Remote Direct Memory Access (RDMA))

2.1.2.3 Special Signals

The AMS GRA Model defines a tailored set of OMS Special Signals for use on AMS GRA Mission Systems. For more information see the Mission System Volume Section 2.4.

2.2 Compute Environment

2.2.1 DPP Compute Environment

The DPP Compute Environment is the collection of Processing Components and infrastructure software that together provide the environment for hosting DPs on a DPP.

The DPP Compute Environment provides a hardware-specific low-level software stack, which includes:

- Host operating systems
- Hypervisors
- Board Support Packages (BSPs)
- DPP API definition and implementation

The DPP Compute Environment provides hardware-agnostic portable infrastructure software (listed below), as notionally depicted in **Figure 2.2.1**.

- Guest Operating Systems
- Container Frameworks
- Containers
- MASI

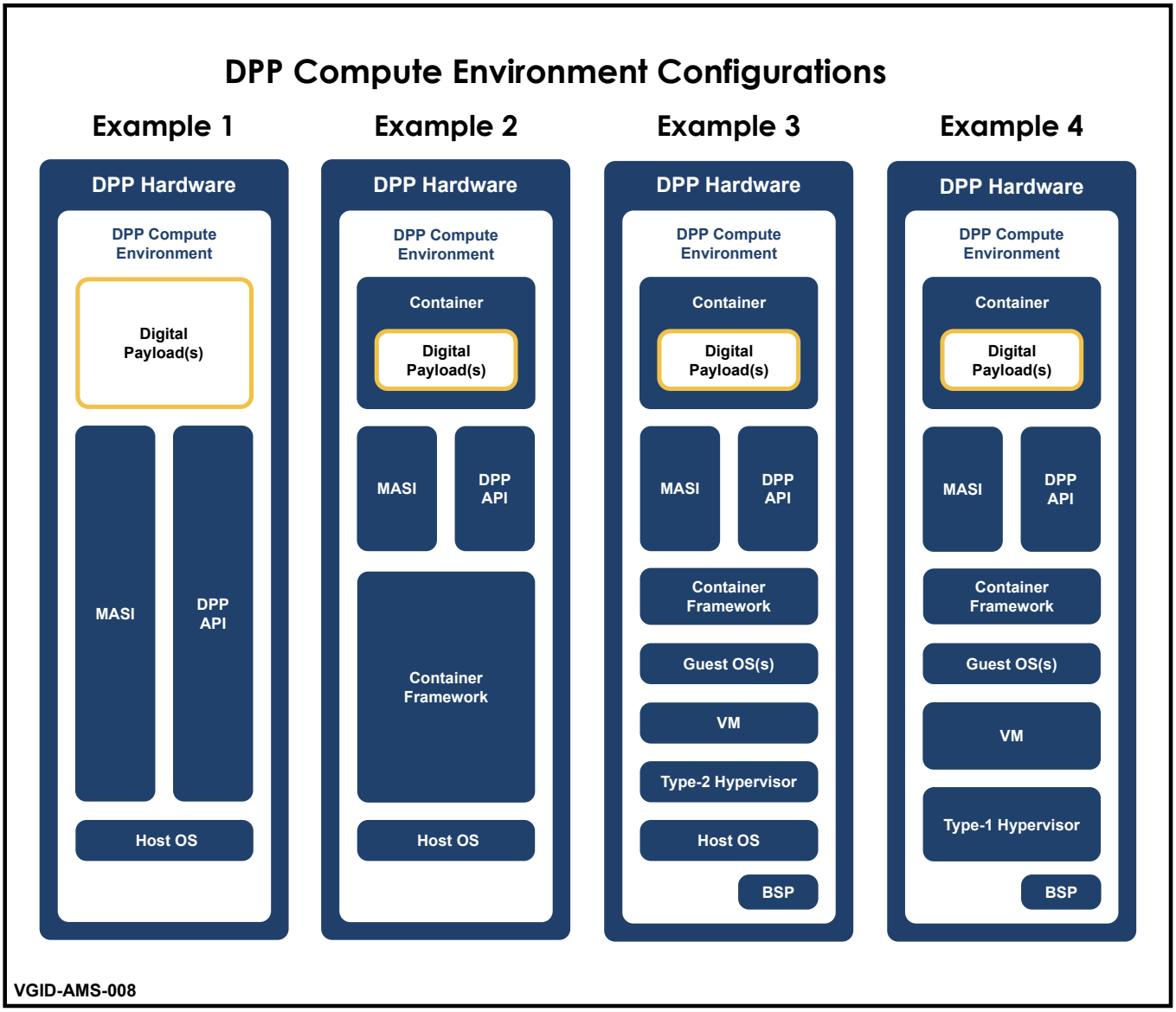


Figure 2.2.1: The AMS GRA Software Architecture Allows for Multiple Possible Deployment Configurations of Skills in the DPP Compute Environment, Allowing Adopters to Choose the Configuration that Best Aligns with Their Mission or Platform Needs.

MASI is the set of Services responsible for tasks such as startup, health assessments, deployment, and logging. These Services support the operation of Skills. See the MASI Volume for more details.

2.2.2 Multi-Function Aperture (MFA) Compute

MFA Compute is the set of processing resources (i.e., GPPs, GPUs, and FPGAs) associated with an MFA. MFA Compute is described in the MFA MPU Volume.

2.3 Development Kit (DK)

DKs enable third-party Skill developers to implement AMS GRA Services, OMS Services, and MFA Local Functions for deployment to DPP and MFA Compute Environments. A DK is different than the Starter Kit (SK). The SK is an implementation of the AMS GRA Software Framework and supporting tools to accelerate software development, integration, and deployment onto an AMS GRA Mission System instantiation. DKs support the development and packing of Level 1 MPUs for their specific Compute Environment. See the DPP and MFA MPU Volumes for details on DKs.

3.0 Compliance

The MSI is responsible for implementing the architectures that realize the software layers discussed above; see the Assembly and Missions Systems Volumes for further information. The MSI is also responsible for implementing the MASI; see the MASI Volume for further information.

An MPU that is delivered in a container is allowed to be removed from the container for execution. Services may also be delivered in a non-containerized format in addition to delivery in a containerized format. Note that GRA-MASI-021 and GRA-MASI-022 do not preclude a MASI from deploying non-containerized software. A procuring entity may remove MPU software from its container if appropriate for their use case. Suppliers may also deliver services in a non-containerized format to support such use cases, so long as they provide a containerized version.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
BSP	Board Support Package
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
COTS	Commercial-Off-The-Shelf
DevSecOps	Development, Security, and Operations
DK	Development Kit
DP	Digital Payload
DPP	Digital Payload Processor
FPGA	Field-Programmable Gate Array
GPP	General Purpose Processor
GPU	Graphics Processing Unit
GRA	Government Reference Architecture
IDD	Interface Description Document
MASI	Mission Agnostic Service Infrastructure
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MPS	Mission Processing Subsystem
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
OMS	Open Mission Systems
OS	Operating System
OSS	Open Source Software
PCaB	Processor Chassis and Backplane
POSIX	Portable Operating System Interface

RDMA	Remote Direct Memory Access
SK	Starter Kit

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the SK continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision did not add any significant changes.	July 24, 2025	SK13.0
B	This revision modifies the usage of POSIX calls for the GPP API, and includes additional details about containerization.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Software Architecture Volume Between Documentation Releases.